```dart
import 'dart:io';

/*
 Task 2 - Question 1

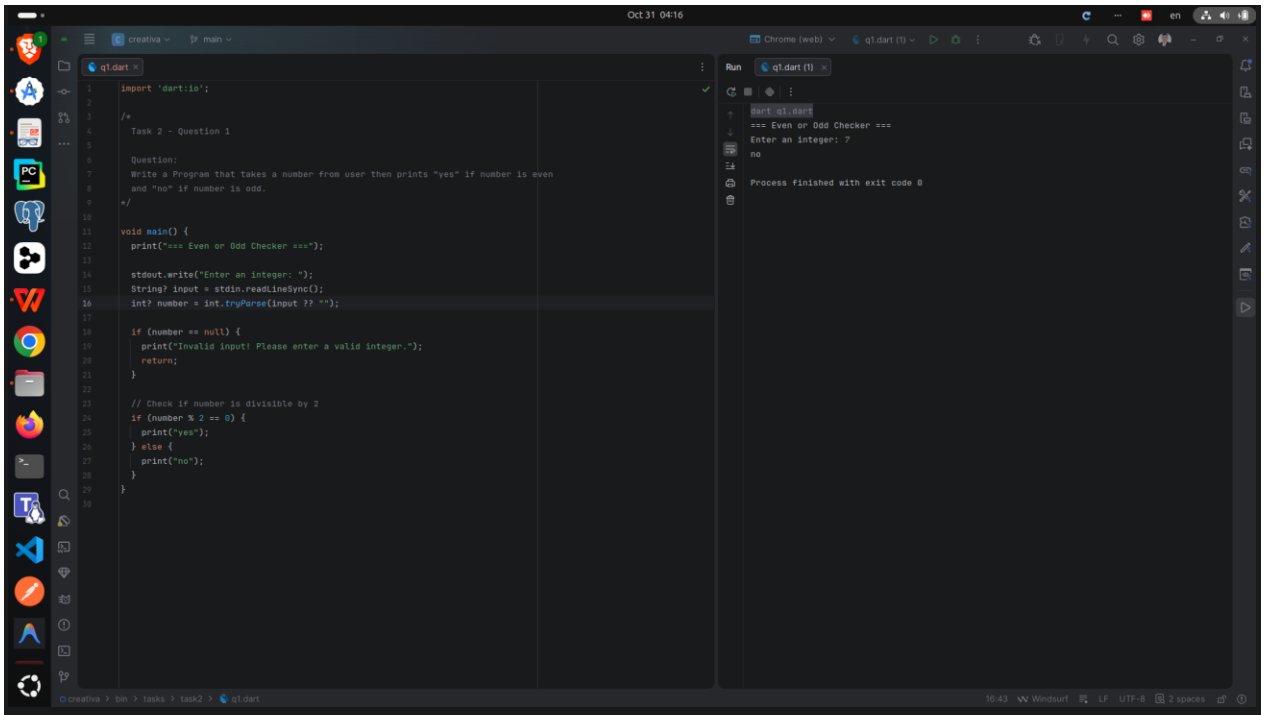
 Question:
 Write a Program that takes a number from user then prints “yes” if number is even
 and “no” if number is odd.
*/

void main() {
 print("=== Even or Odd Checker ===");

 stdout.write("Enter an integer: ");
 String? input = stdin.readLineSync();
 int? number = int.tryParse(input ?? "");

 if (number == null) {
  print("Invalid input! Please enter a valid integer.");
  return;
 }

 // Check if number is divisible by 2
 if (number % 2 == 0) {
  print("yes");
 } else {
  print("no");
 }
}
```




```dart
import 'dart:io';

/*
 Task 2 - Question 2

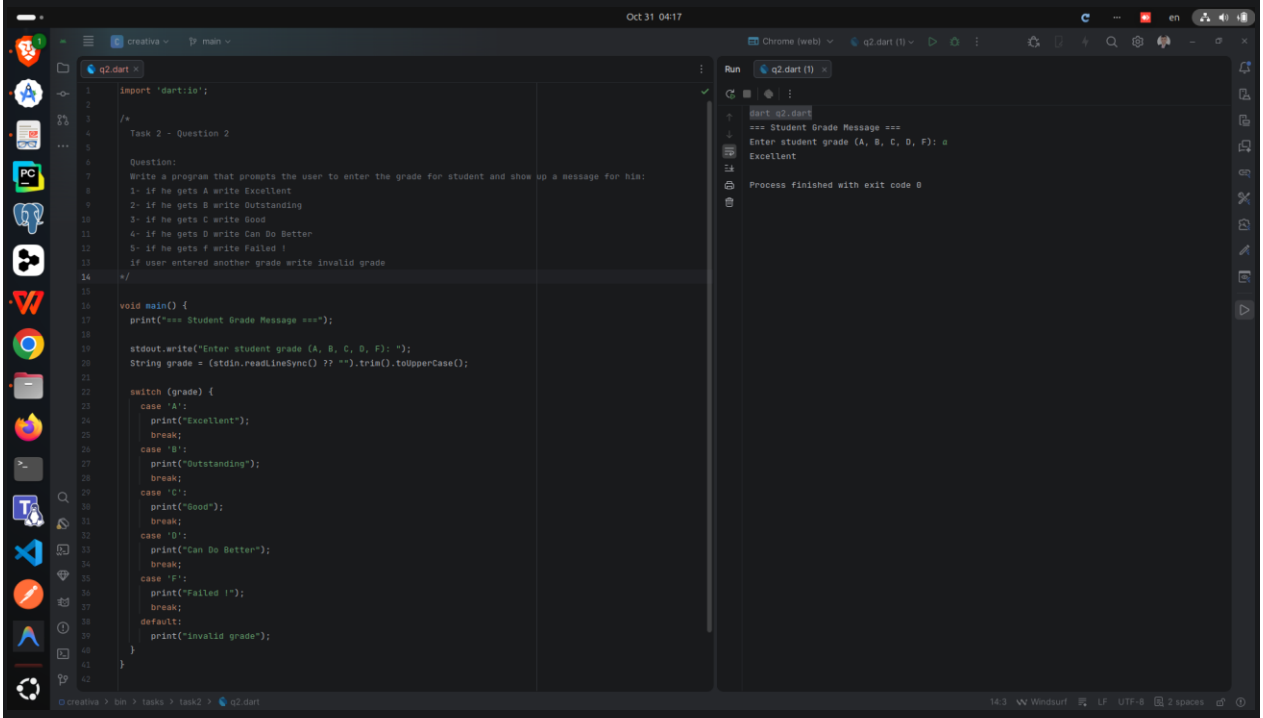
 Question:
 Write a program that prompts the user to enter the grade for student and show up a message for him:
 1- if he gets A write Excellent
 2- if he gets B write Outstanding
 3- if he gets C write Good
 4- if he gets D write Can Do Better
 5- if he gets f write Failed !
 if user entered another grade write invalid grade
*/

void main() {
 print("=== Student Grade Message ===");

 stdout.write("Enter student grade (A, B, C, D, F): ");
 String grade = (stdin.readLineSync() ?? "").trim().toUpperCase();

 switch (grade) {
  case 'A':
   print("Excellent");
   break;
  case 'B':
```

```
    print("Outstanding");
    break;
  case 'C':
    print("Good");
    break;
  case 'D':
    print("Can Do Better");
    break;
  case 'F':
    print("Failed !");
    break;
  default:
    print("invalid grade");
 }
}
```




```dart
import 'dart:io';

/*
 Task 2 - Question 3: Student Report Card

 Question:
 A student has two test scores (double). Calculate the average,
 check if they passed (>= 50),
 then print their grade using else if:
```

```
 90+     → A
 80–89   → B
 70–79   → C
 60–69   → D
 below 60 → F
*/

void main() {
 print("=== Student Report Card ===");

 stdout.write("Enter first test score: ");
 double? score1 = double.tryParse(stdin.readLineSync() ?? "");

 stdout.write("Enter second test score: ");
 double? score2 = double.tryParse(stdin.readLineSync() ?? "");

 if (score1 == null || score2 == null) {
  print("Invalid input! Please enter valid numeric scores.");
  return;
 }

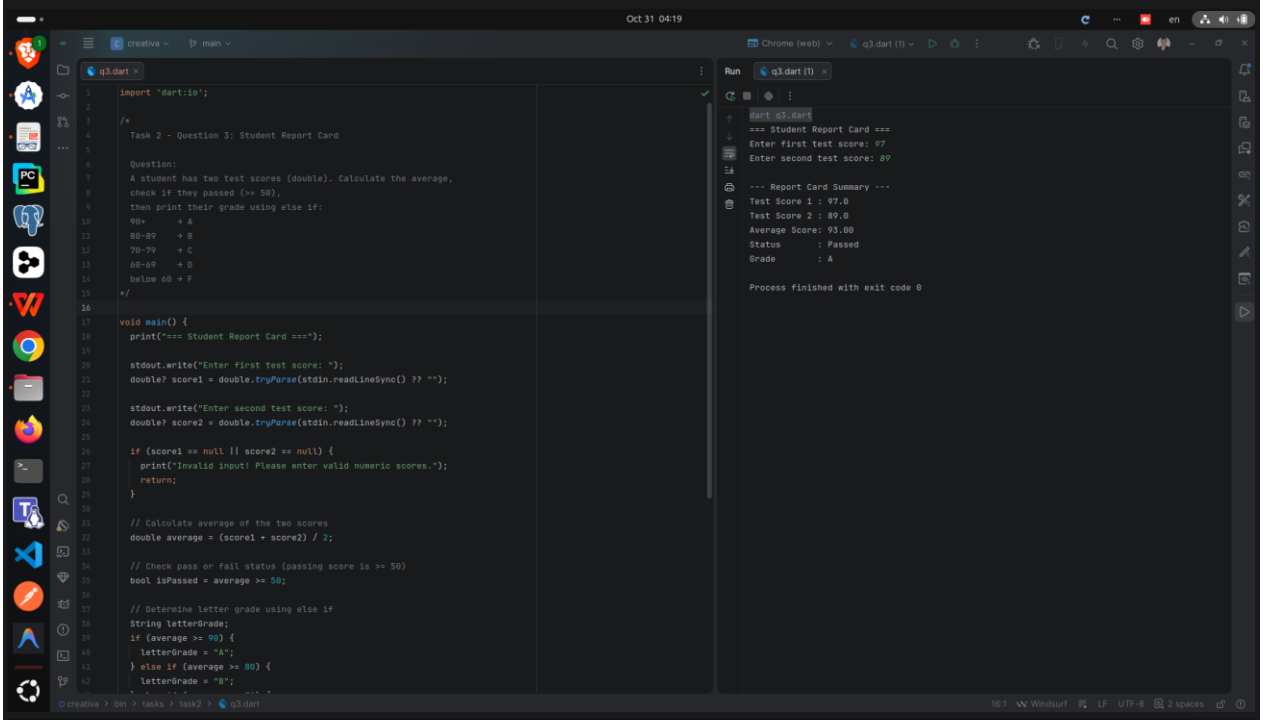
 // Calculate average of the two scores
 double average = (score1 + score2) / 2;

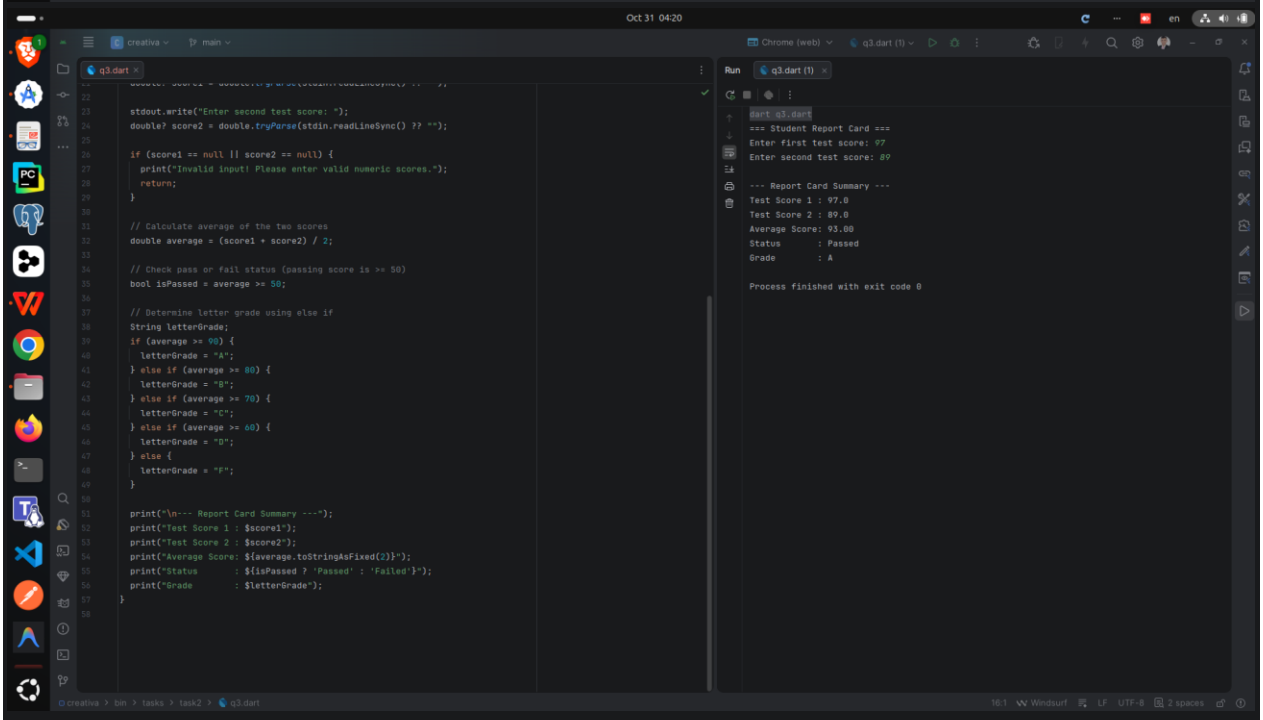
 // Check pass or fail status (passing score is >= 50)
 bool isPassed = average >= 50;

 // Determine letter grade using else if
 String letterGrade;
 if (average >= 90) {
  letterGrade = "A";
 } else if (average >= 80) {
  letterGrade = "B";
 } else if (average >= 70) {
  letterGrade = "C";
 } else if (average >= 60) {
  letterGrade = "D";
 } else {
  letterGrade = "F";
 }

 print("\n--- Report Card Summary ---");
 print("Test Score 1 : $score1");
 print("Test Score 2 : $score2");
 print("Average Score: ${average.toStringAsFixed(2)}");
 print("Status      : ${isPassed ? 'Passed' : 'Failed'}");
 print("Grade       : $letterGrade");
```

```
}
```






```
/*
Task 2 - Question 4: Code Tracing
```

Question:
Read the code below carefully, then do a Tracing for the Steps that the code goes through.

Given Code:

```
1  void main() {
2    int number = 14;
3
4    if (number % 2 8== 0) {
5      print("Even number");
6    } else {
7      print("Odd number");
8    }
9  }
```

==========
Step-by-Step Code Tracing:
==========
Step 1 (Line 1):
 - `void main()`
 - The program entry point is invoked.

Step 2 (Line 2):
 - `int number = 14;`
 - A local integer variable `number` is allocated in memory and initialized to `14`.
 - Memory state: { number: 14 }

Step 3 (Line 4):
 - `if (number % 2 == 0)`
 - The expression is evaluated:
   * `number % 2` -> `14 % 2` = 0 (14 divided by 2 gives 7 with remainder 0).
   * Next, `0 == 0` is evaluated.
   * The condition returns boolean `true`.

Step 4 (Line 5):
 - `print("Even number");`
 - Because the condition evaluated to `true`, control enters the `if` body block.
 - The text "Even number" is printed to the console output.

Step 5 (Lines 6-8):
 - `else { print("Odd number"); }`
 - Since the `if` condition was met, the `else` block is completely skipped.

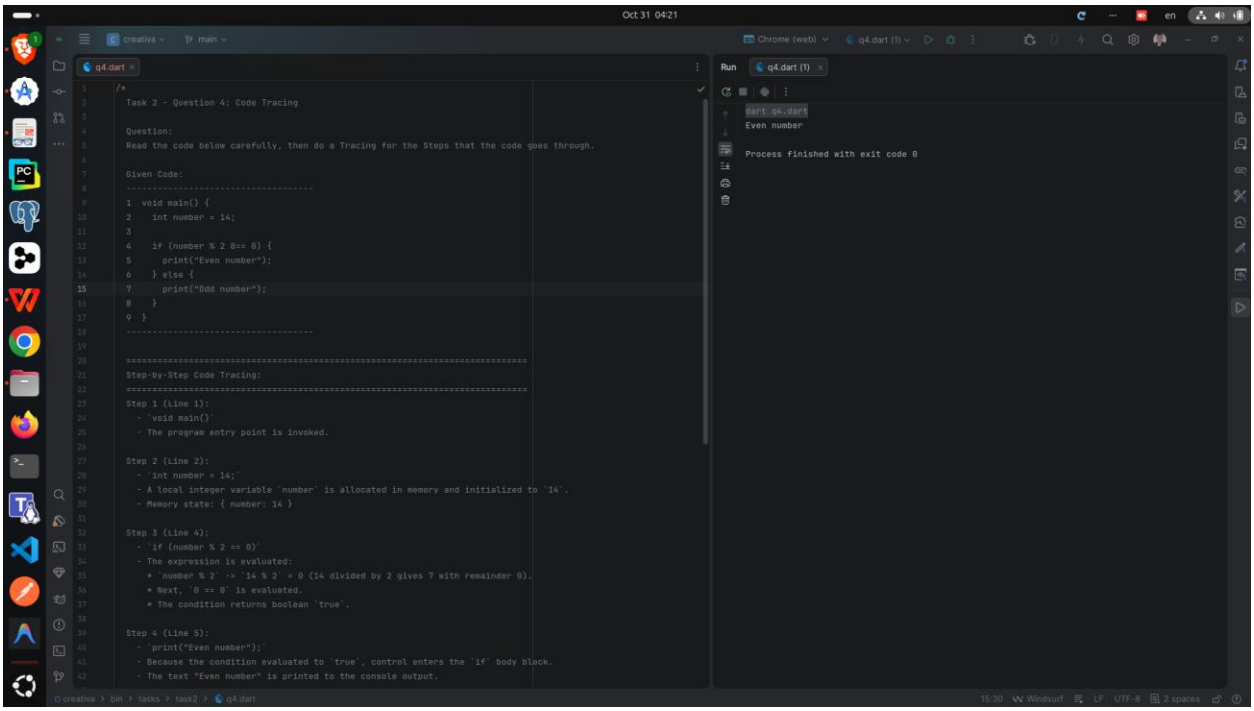
Step 6 (Line 9):
 - `}`

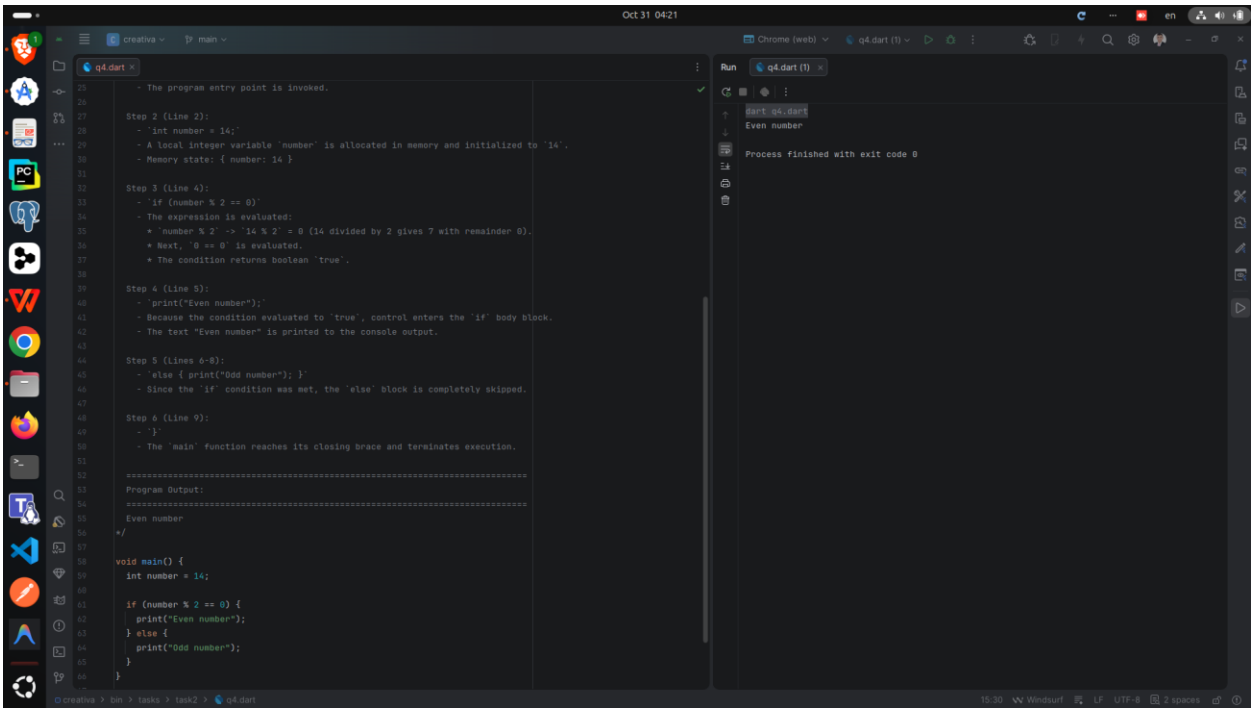
```
   - The `main` function reaches its closing brace and terminates execution.


  ================================================================================
  Program Output:
  ================================================================================
  Even number
*/

void main() {
  int number = 14;

  if (number % 2 == 0) {
    print("Even number");
  } else {
    print("Odd number");
  }
}
```


Oct 31 04:21
creativa
main
Chrome (web)
q4.dart (1)
q4.dart
Run
dart q4.dart
Even number
Process finished with exit code 0


Oct 31 04:21
creativa
main
Chrome (web)
q4.dart (1)
q4.dart
Run
dart q4.dart
Even number
Process finished with exit code 0